

Assignment 8: Time Series Analysis

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A08_TimeSeries.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Check your working directory
 - Load the tidyverse, lubridate, zoo, and trend packages
 - Set your ggplot theme

```
# load packages
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.3      v tidyr     1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(lubridate)
library(trend)
library(zoo)
```

```
##
## Attaching package: 'zoo'
##
## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric
```

```
library(here)
```

```
## here() starts at /home/guest/EDE_Fall2025
```

```
# check working directory
here
```

```
## function (...)
## {
##     .root_env$root$f(...)
## }
## <bytecode: 0x5747b38fb7b0>
## <environment: namespace:here>
```

```
# set ggplot theme
mytheme <- theme_classic() +
  theme(
    # set legend position
    legend.position = "bottom",

    # set fonts
    plot.title = element_text(size = 14),
    axis.title = element_text(size = 12),
    axis.text = element_text(size = 10, color = "black"),
    legend.text = element_text(size = 10, color = "black")
  )
theme_set(mytheme)
```

2. Import the ten datasets from the Ozone_TimeSeries folder in the Raw data folder. These contain ozone concentrations at Garinger High School in North Carolina from 2010-2019 (the EPA air database only allows downloads for one year at a time). Import these either individually or in bulk and then combine them into a single dataframe named **GaringerOzone** of 3589 observation and 20 variables.

```
#1
ozone_files <- list.files(here("Data/Raw/Ozone_TimeSeries/"),
                          full.names = TRUE)

# import all files and combine into a single df
GaringerOzone <- map(ozone_files,
                     read.csv) %>% bind_rows()
```

Wrangle

3. Set your date column as a date class.

4. Wrangle your dataset so that it only contains the columns Date, Daily.Max.8.hour.Ozone.Concentration, and DAILY_AQI_VALUE.
5. Notice there are a few days in each year that are missing ozone concentrations. We want to generate a daily dataset, so we will need to fill in any missing days with NA. Create a new data frame that contains a sequence of dates from 2010-01-01 to 2019-12-31 (hint: `as.data.frame(seq())`). Call this new data frame Days. Rename the column name in Days to “Date”.
6. Use a `left_join` to combine the data frames. Specify the correct order of data frames within this function so that the final dimensions are 3652 rows and 3 columns. Call your combined data frame GaringerOzone.

```
# 3
# set date column as a Date
GaringerOzone$Date <- mdy(GaringerOzone$Date)

# 4
# select relevant columns
GaringerOzone <- GaringerOzone %>%
  select(Date, Daily.Max.8.hour.Ozone.Concentration, DAILY_AQI_VALUE)

# 5
# create df of all dates in range
Days <- tibble(
  "Date" = seq(as.Date("2010-01-01"), as.Date("2019-12-31"), by = "day")
)

# 6
# left join dfs to get observations for all dates in range
GaringerOzone <- Days %>%
  left_join(GaringerOzone, by = "Date")

summary(GaringerOzone) # see 63 days with missing data
```

```
##      Date      Daily.Max.8.hour.Ozone.Concentration DAILY_AQI_VALUE
## Min.   :2010-01-01      Min.   :0.00200              Min.    : 2.00
## 1st Qu.:2012-07-01      1st Qu.:0.03200              1st Qu.: 30.00
## Median :2014-12-31      Median :0.04100              Median : 38.00
## Mean   :2014-12-31      Mean   :0.04163              Mean    : 41.57
## 3rd Qu.:2017-07-01      3rd Qu.:0.05100              3rd Qu.: 47.00
## Max.   :2019-12-31      Max.   :0.09300              Max.    :169.00
##      NA's      :63
```

Visualize

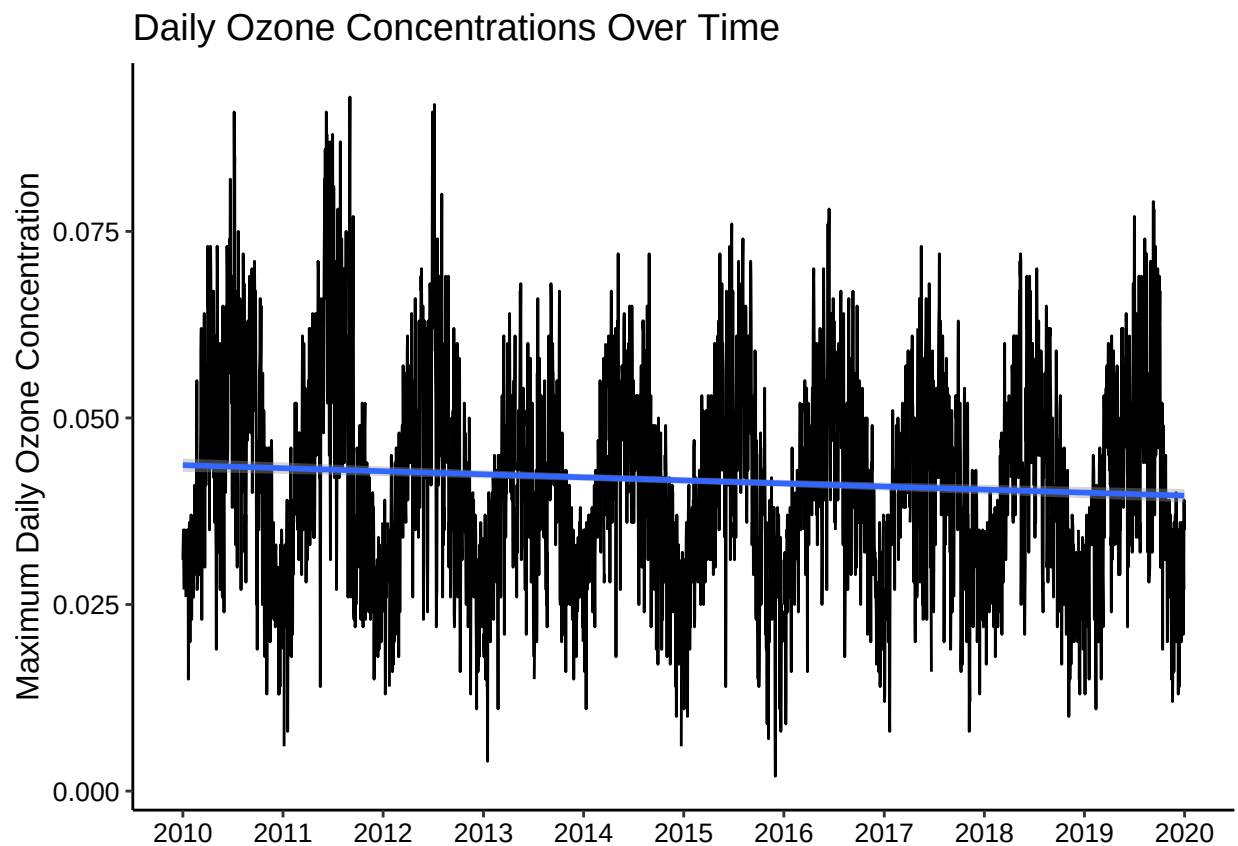
7. Create a line plot depicting ozone concentrations over time. In this case, we will plot actual concentrations in ppm, not AQI values. Format your axes accordingly. Add a smoothed line showing any linear trend of your data. Does your plot suggest a trend in ozone concentration over time?

```
#7
# plot of ozone concentrations over time
ggplot(GaringerOzone, aes(x = Date, y = Daily.Max.8.hour.Ozone.Concentration)) +
  geom_line() +
  geom_smooth(method = "lm") +
```

```
scale_x_date(date_breaks = "1 year", date_labels = "%Y") +
labs(
  x = NULL,
  y = "Maximum Daily Ozone Concentration",
  title = "Daily Ozone Concentrations Over Time"
)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 63 rows containing non-finite outside the scale range
## ('stat_smooth()').
```



Answer: This plot suggests there is a very slight negative trend in ozone concentrations over time.

Time Series Analysis

Study question: Have ozone concentrations changed over the 2010s at this station?

8. Use a linear interpolation to fill in missing daily data for ozone concentration. Why didn't we use a piecewise constant or spline interpolation?

#8

```
GaringerOzone <- GaringerOzone %>%  
  mutate(Daily.Max.8.hour.Ozone.Concentration =  
    zoo::na.approx(Daily.Max.8.hour.Ozone.Concentration))
```

Answer: Linear interpolation can be used when there are small periods of missing data in your dataset as it's used to "connect the dots" between the existing data points. Piecewise interpolation uses the nearest neighbor to fill in missing data. This method would not work well with the data because the seasonal component means the data is constantly fluctuating, so assuming that a data point would be equal to its nearest neighbor would not be a good representation of the movement of the data. A spline interpolation functions similarly to the linear interpolation but assumes the data follows a quadratic function. The data does not appear to follow a quadratic function so this would not be a good approximation of the missing data.

9. Create a new data frame called `GaringerOzone.monthly` that contains aggregated data: mean ozone concentrations for each month. In your pipe, you will need to first add columns for year and month to form the groupings. In a separate line of code, create a new Date column with each month-year combination being set as the first day of the month (this is for graphing purposes only)

#9

```
GaringerOzone.monthly <- GaringerOzone %>%  
  mutate(month = month(Date),  
         year = year(Date)) %>%  
  group_by(month, year) %>%  
  summarize(mean_ozone_concentration =  
    mean(Daily.Max.8.hour.Ozone.Concentration)) %>%  
  ungroup() %>%  
  mutate(Date = my(paste(month, year, sep = "-"))) %>%  
  relocate(Date) %>%  
  arrange(Date)
```

'summarise()' has grouped output by 'month'. You can override using the
'.groups' argument.

10. Generate two time series objects. Name the first `GaringerOzone.daily.ts` and base it on the dataframe of daily observations. Name the second `GaringerOzone.monthly.ts` and base it on the monthly average ozone values. Be sure that each specifies the correct start and end dates and the frequency of the time series.

#10

```
first_day <- day(first(GaringerOzone$Date))  
first_month <- month(first(GaringerOzone$Date))  
first_year <- year(first(GaringerOzone$Date))  
  
last_day <- day(last(GaringerOzone$Date))  
last_month <- month(last(GaringerOzone$Date))  
last_year <- year(last(GaringerOzone$Date))  
  
# Note: setting ending day as c(last_year, last_month) ends up getting the  
# incorrect observations for the daily time series, so I've excluded the end
```

```

GaringerOzone.daily.ts <- ts(GaringerOzone$Daily.Max.8.hour.Ozone.Concentration,
                             start = c(first_year, first_day),
                             frequency = 365)

GaringerOzone.monthly.ts <- ts(GaringerOzone.monthly$mean_ozone_concentration,
                                start = c(first_year, first_month),
                                end = c(last_year, last_month),
                                frequency = 12)

```

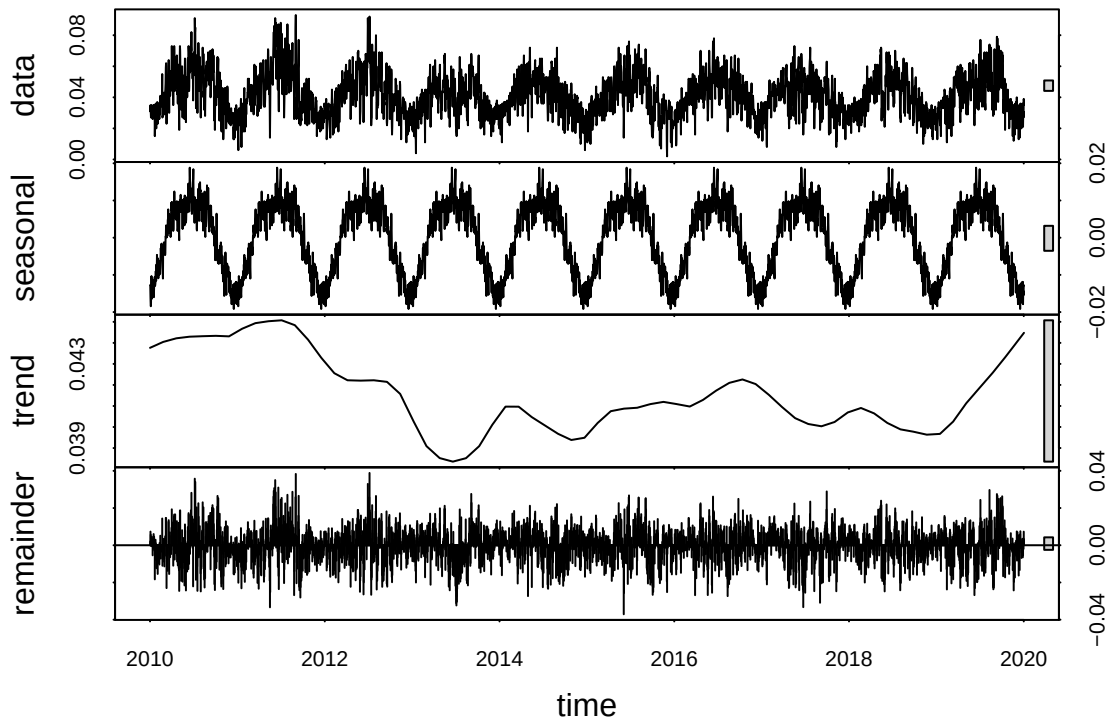
11. Decompose the daily and the monthly time series objects and plot the components using the `plot()` function.

#11

```

GaringerOzone.daily.decomposed <- stl(GaringerOzone.daily.ts,
                                       s.window = "periodic")
plot(GaringerOzone.daily.decomposed)

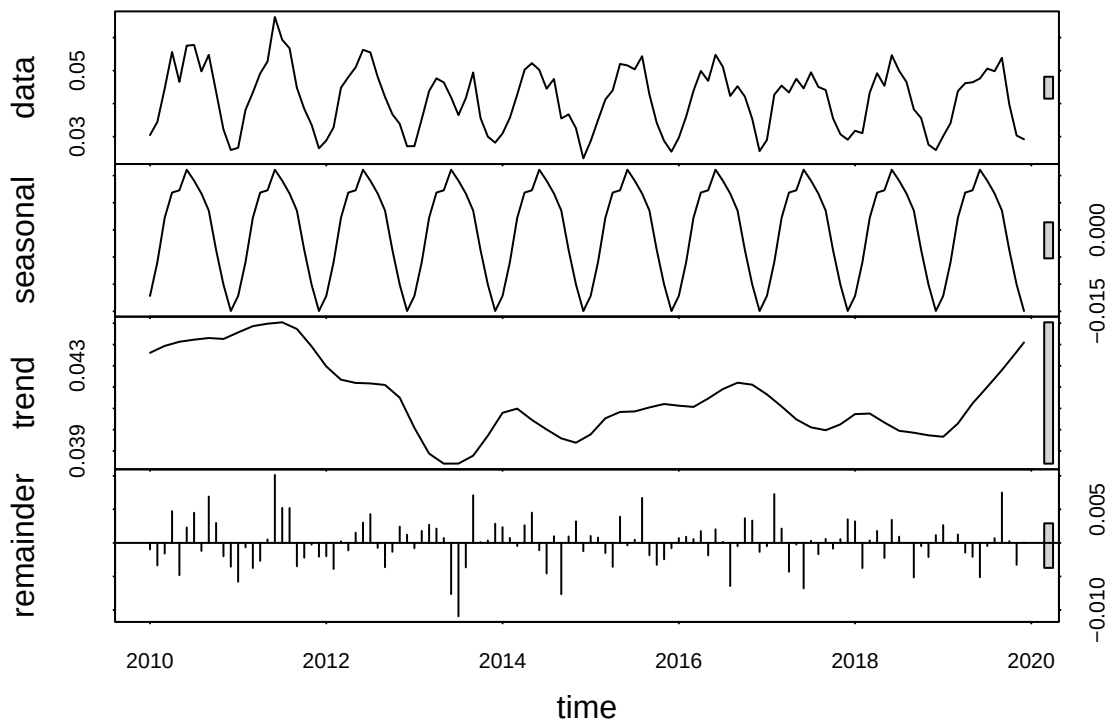
```



```

GaringerOzone.monthly.decomposed <- stl(GaringerOzone.monthly.ts,
                                          s.window = "periodic")
plot(GaringerOzone.monthly.decomposed)

```



12. Run a monotonic trend analysis for the monthly Ozone series. In this case the seasonal Mann-Kendall is most appropriate; why is this?

#12

```
monthly_ozone_trend <- trend::smk.test(GaringerOzone.monthly.ts)
monthly_ozone_trend      # p-value = 0.04965
```

```
##
## Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: GaringerOzone.monthly.ts
## z = -1.963, p-value = 0.04965
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##      S varS
## -77 1499
```

```
summary(monthly_ozone_trend)
```

```
##
## Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: GaringerOzone.monthly.ts
```

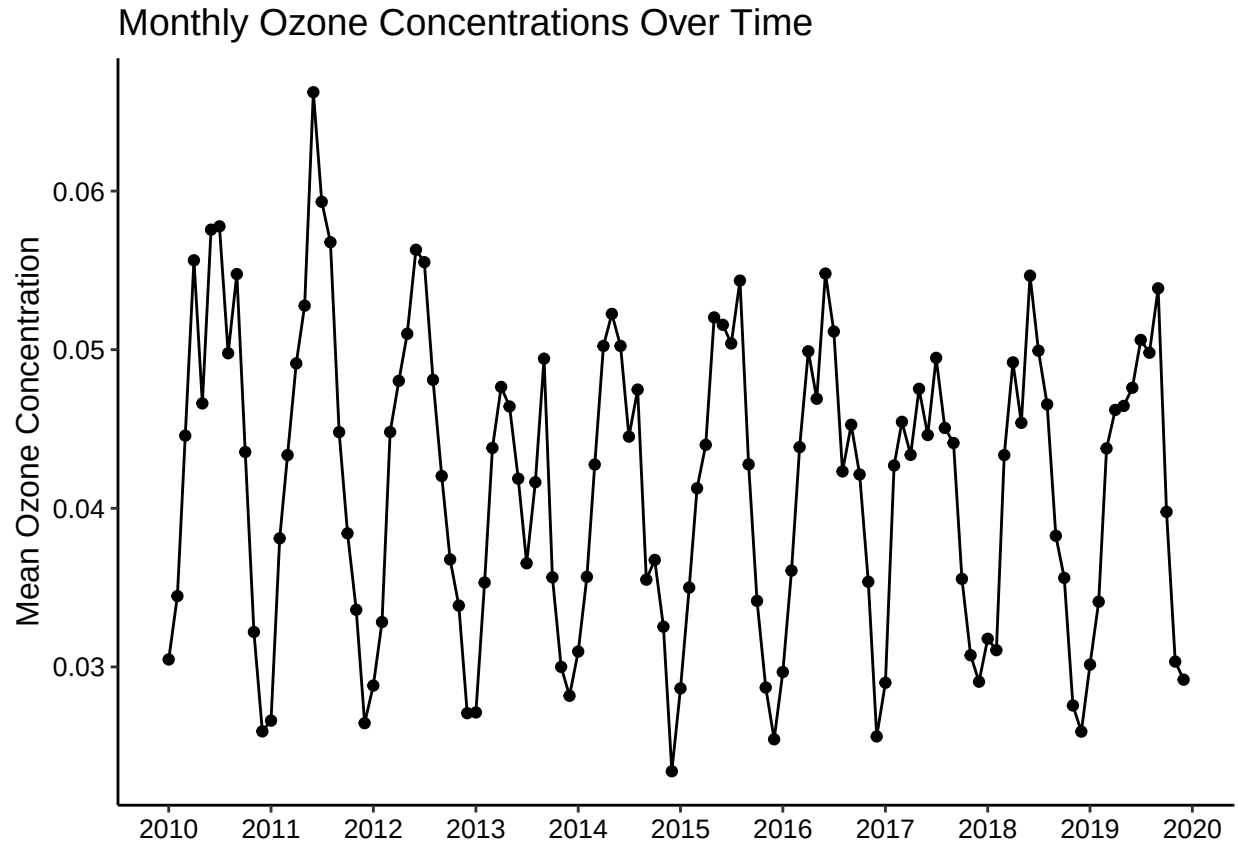
```
## alternative hypothesis: two.sided
##
## Statistics for individual seasons
##
## H0
##
##      S varS    tau      z Pr(>|z|)
## Season 1:  S = 0   15  125  0.333  1.252  0.21050
## Season 2:  S = 0   -1  125 -0.022  0.000  1.00000
## Season 3:  S = 0   -4  124 -0.090 -0.269  0.78762
## Season 4:  S = 0  -17  125 -0.378 -1.431  0.15241
## Season 5:  S = 0  -15  125 -0.333 -1.252  0.21050
## Season 6:  S = 0  -17  125 -0.378 -1.431  0.15241
## Season 7:  S = 0  -11  125 -0.244 -0.894  0.37109
## Season 8:  S = 0   -7  125 -0.156 -0.537  0.59151
## Season 9:  S = 0   -5  125 -0.111 -0.358  0.72051
## Season 10: S = 0  -13  125 -0.289 -1.073  0.28313
## Season 11: S = 0  -13  125 -0.289 -1.073  0.28313
## Season 12: S = 0   11  125  0.244  0.894  0.37109
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Answer: The seasonal Mann-Kendall is most appropriate for this series because it has a clear seasonal component, and the SMK test can be used to analyze the trend of a time series with seasonality.

13. Create a plot depicting mean monthly ozone concentrations over time, with both a `geom_point` and a `geom_line` layer. Edit your axis labels accordingly.

```
# 13

ggplot(GaringerOzone.monthly, aes(x = Date, y = mean_ozone_concentration)) +
  geom_line() +
  geom_point() +
  scale_x_date(date_breaks = "year",
               date_labels = "%Y") +
  labs(x = NULL,
       y = "Mean Ozone Concentration",
       title = "Monthly Ozone Concentrations Over Time")
```

14. To accompany your graph, summarize your results in context of the research question. Include output from the statistical test in parentheses at the end of your sentence. Feel free to use multiple sentences in your interpretation.

Answer: This graph shows that there is a clear seasonality in ozone concentrations over time, with levels peaking during the summer. There is not a clear monotonic trend that is visually evident in this graph. The Seasonal Mann-Kendall test found that there is a significant monotonic trend at a 95% confidence interval, though these results are not very strong ($p\text{-value} = 0.04965 < 0.05$).

15. Subtract the seasonal component from the `GaringerOzone.monthly.ts`. Hint: Look at how we extracted the series components for the `EnoDischarge` on the lesson Rmd file.
16. Run the Mann Kendall test on the non-seasonal Ozone monthly series. Compare the results with the ones obtained with the Seasonal Mann Kendall on the complete series.

#15

```
GaringerOzone.monthly.nonseasonal.ts <- GaringerOzone.monthly.ts -
  GaringerOzone.monthly.decomposed$time.series[, "seasonal"]

# # visualize observations without seasonal component
# GaringerOzone.monthly.nonseasonal <- GaringerOzone.monthly.nonseasonal.ts %>%
#   as.data.frame() %>%
#   rename("observed_minus_seasonal" = x) %>%
#   mutate(Date = GaringerOzone.monthly$Date,
```

```

#           observed = GaringerOzone.monthly$mean_ozone_concentration) %>%
#   relocate(Date)
#
# GaringerOzone.monthly.nonseasonal %>%
#   pivot_longer(cols = c("observed", "observed_minus_seasonal"),
#                 values_to = "mean_ozone_concentration",
#                 names_to = "component") %>%
#   ggplot(aes(x = Date, y = mean_ozone_concentration, color = component)) +
#   geom_line() +
#   scale_color_manual(values = c("darkgrey", "red"))

#16

monthly_ozone_trend_nonseasonal <- trend::mk.test(GaringerOzone.monthly.nonseasonal.ts)
monthly_ozone_trend_nonseasonal

##
## Mann-Kendall trend test
##
## data: GaringerOzone.monthly.nonseasonal.ts
## z = -2.672, n = 120, p-value = 0.00754
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##           S           varS           tau
## -1.179000e+03  1.943657e+05 -1.651376e-01

summary(monthly_ozone_trend_nonseasonal)

##           Length Class  Mode
## data.name      1    -none- character
## p.value         1    -none- numeric
## statistic       1    -none- numeric
## null.value      1    -none- numeric
## parameter       1    -none- numeric
## estimates       3    -none- numeric
## alternative     1    -none- character
## method          1    -none- character
## pvalg           1    -none- numeric

```

Answer: The p-value of the Mann-Kendall test run on the series after removing the seasonal component is 0.00754. These results are significant at a 99% confidence interval and allow us to reject our null hypothesis that this series is stationary. These results are more significant than the results of the SMK test on the data that included the seasonal component.